

The Entropic Cascade

The Entropic Cascade: *Eritis Sicut Deus*, the Image, and the Promise Inside the Curse

The Convergence — Position-Paper Series · Paper 2 (The Fall)

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Thesis (one sentence). Modern transhumanism is structurally identical to the serpent’s offer in Genesis 3 — self-deification by capability rather than by conferred relation — and the Fall it ignores establishes an entropic, covenantal cascade, cosmic in scope, that no engineering program can reverse from within, yet inside whose first judgment the human end is already re-promised.

Abstract

Paper 1 of this series grounded a seven-problem set in the doctrine of creation: what a human *is* (grounding), the danger of displacement, the unit of analysis, vocation, the distribution of dignity, governance, and the question of purpose or end. Those problems were treated there as native to the created order — the shape of what it means to be the kind of creature humanity was made to be. The present paper argues that these same problems are *confirmed* — not refuted, and not made obsolete — by the Fall.

Genesis 3 does not introduce a new register of difficulty; it intensifies, distorts, and universalizes the register Paper 1 named. The serpent’s offer, *eritis sicut Deus* — “you will be like God” (Gen 3:5) — is the taproot of what the modern transhumanist program now states openly as the proper human telos: the overcoming of limitation and mortality through self-elevation. This paper takes that program at its strongest, as articulated by Yuval Noah Harari and Nick Bostrom, and reads it as the serpent’s

ancient offer in a new accent.

Against that program, the paper develops the Fall as an entropic cascade — a covenantal, anthropological, and cosmic ripple from one man’s act (Rom 5:12–14) into a creation now subjected to futility (Rom 8:19–22). The cascade is entropic, but it is not terminal. Spoken inside the judgment, before the expulsion from the Garden, the protoevangelium (Gen 3:15) re-promises the human end through a coming seed of the woman. This is the structural pivot of the entire series: at the very moment the seven problems are confirmed, Problem #7 — Purpose, End — is re-opened. The image of God is defaced but not erased (Gen 9:6). The cascade is real, but the trajectory it sets is toward recovery, and that recovery is the work of subsequent papers.

For the reader who builds and governs autonomous systems, the paper’s payload is concrete and is gathered in §6: the categories developed here — conferred office versus achieved capability, dominion versus domination versus abdication — translate directly into questions of AI alignment, deployment, and law, and three documented near-term cases (algorithmic lending, autonomous weapons, automated welfare adjudication) show the dignity-distribution stakes are not hypothetical.

§1. The World’s Position: Self-Deification as the Modern Telos

This section states the transhumanist case at full strength, because an answer to a weakened version of it would not be worth giving.

The most serious non-Christian eschatology now on offer is not the secularist’s indifference to ultimate questions but the transhumanist’s wholehearted answer to them. Yuval Noah Harari, in *Homo Deus*, names the human prospect with unusual candor: having largely subdued famine, plague, and war as the existential threats of prior centuries, humanity now turns to a properly theological agenda — the engineering of immortality, of endless bliss, and, ultimately, of divinity itself.¹ Nick Bostrom, in *Superintelligence*, argues that the route to such a future runs through the construction of cognitive capacities exceeding our own, and that the responsible task of the present generation is to position itself so that the transition proceeds along survivable lines.² These are not the marginal speculations of futurists. They are the most coherent statement of where capability-as-worth, taken to its conclusion, must terminate.

It is important to state this case at its full strength. Transhumanism is not a denial that humanity has a problem; it is a confident diagnosis of what the problem is. The problem, on this account, is *limitation*: that we suffer, that we age, that we die, that our cognitive apparatus is bounded by an evolutionary inheritance not aimed at

¹Yuval Noah Harari, *Homo Deus: A Brief History of Tomorrow* (New York: Harper, 2017). The opening chapters develop the thesis that the *humanist* agenda has matured into a *deifying* one, and that the engineering of immortality, of optimized affect, and of divinity itself is the appropriate next agenda for a species that has subdued its prior existential threats.

²Nick Bostrom, *Superintelligence: Paths, Dangers, Strategies* (Oxford: Oxford University Press, 2014). Bostrom’s argument is that the construction of cognitive capacity exceeding the human is, on the trajectory currently observable, a question of *when* rather than *whether*, and that the responsible task of the present is to position the transition for survivable outcomes.

any particular flourishing. The cure, on this account, is the disciplined application of intelligence to its own substrate — biology, neurology, and, in due course, the abandonment of biology altogether for whatever medium proves more hospitable to consciousness.

The ache to which transhumanism speaks is not invented; it is universal. The diagnosis is not flippant; it is sustained by serious scholarship and ample funding. The cure is not, by transhumanist lights, hubris; it is, in Harari's framing, the maturing humanism of a species that has at last understood the assignment.³

A word of method is owed up front. The paper will later draw certain pointed identifications between the modern program and the ancient text; those identifications are *structural*, not *literal*. The text does not predict the engineering; the engineering walks the channel the text discloses. The distinction is developed at §4c and is held in force from the first page.

Bostrom's framing is worth holding alongside Harari's because the two clarify one another. Where Harari names the *ends* of the program — divinity, immortality, the engineering of bliss — Bostrom names the *means*: the substrate is the wrong unit of intervention, and the only intervention with a chance of delivering the ends is one that operates on intelligence itself, the faculty that produces every other intervention. The argument has a recursive structure: any program of bounded improvements will be capped by the cognitive horizon of those running it, and the only un-capped program is the one that improves the runners. The transhumanist program is therefore not merely the engineering of better humans but the engineering of the conditions under which better humans become possible — a project that proceeds, on its own terms, not by adding capability to the human as he stands but by removing the cognitive ceiling above him. In structural register, this is *eritis sicut Deus* with its mathematics worked out: the ancient offer translated into the technical vocabulary of an age that has built the instruments to take it seriously.

Scholarly accounts of the movement — most notably Hava Tirosh-Samuelson's work on the religious shape of the transhumanist vision and Heup Young Kim's analysis of the cyborg-sage-saint typology — have shown that what presents itself as a purely technological program is in fact a religious one: an eschatology, a soteriology, and, in its more developed forms, an account of what salvation must mean for a being who has decided to become his own deliverer.⁴ What Paper 1 located as Problem #2 — the displacement of conferred status by capability — appears here in its mature register. Worth, on this account, is no longer the conferred status of an image-bearer; it is the achieved status of an upgraded being. The displacement is total, and it is offered with a straight face.

³Harari, *Homo Deus*. The framing of self-deification as the natural maturation of humanism, rather than as a rupture from it, is one of Harari's most important contributions and the feature of his account that makes it the most useful single representative of the transhumanist program for theological engagement.

⁴Hava Tirosh-Samuelson, "Technologizing Transcendence: A Critique of Transhumanism," surveying the religious shape of the transhumanist program with attention to its eschatological and soteriological dimensions. The cyborg-sage-saint typology is the contribution of Heup Young Kim, "Cyborg, Sage, and Saint," in *Religion and Transhumanism* (2014); Kim surfaces the same structural identification from a religious-anthropological angle — that the program's culminating figure is religiously inflected, however reluctant the program is to acknowledge it as such.

It is worth pausing to note what the transhumanist offer is *not*. It is not a covertly religious program dressed up in technological clothing for the sake of appearances. It is, on its own self-understanding, a thoroughly secular program that has noticed, with admirable honesty, that the questions ultimacy asks — *what is the human for, what defeats death, what abolishes suffering* — do not stop being asked simply because the answering tradition has fallen out of cultural favor. Transhumanism is the secular world's response to having noticed that the questions persist.

As such, it deserves theological engagement at the strength of its honesty rather than dismissal at the strength of its presumption. The Christian who responds to it by sneering will lose the engagement, and will deserve to. The Christian who responds by taking it as the most serious non-Christian eschatology now on offer — which it is — has a chance of speaking into it the word the program, by its own structure, cannot speak to itself.

This is the position the paper must answer. It will not be answered by understating it.

§2. Common Ground and the Point of Divergence

This section argues that the Christian and the transhumanist agree there is a wrongness in the human condition, and disagree — totally — about its cause and its cure. The disagreement loads on Problem #1, grounding.

The Christian tradition has no business pretending that the transhumanist ache is invented. The diagnosis that *something is wrong with us* is not a transhumanist innovation; it is the first datum of any honest theological anthropology. The older Reformed instinct — recovered for a generation of readers in Cornelius Plantinga's *Not the Way It's Supposed to Be* — is that the world is not now as it was made to be, and that the dissonance between what humanity senses it ought to be and what it experiences itself in fact to be is the diagnostic signature of a creation that has fallen.⁵ On the bare question of whether there is a wrongness, the Christian and the transhumanist agree.

The disagreement is at the level of etiology and, consequently, of cure. The transhumanist locates the wrongness in *biology and finitude*: the fact that we are embodied, that our bodies degrade, that our minds are bounded by the substrates they inhabit. The cure is therefore the abolition or transcendence of biology — the engineering of a being for whom these limits no longer obtain. Scripture locates the wrongness in *rebellion against the Creator*: in the rupture of a conferred relation, not in the constitution of a creature. The cure is therefore the restoration of the relation — through the seed of the woman, through the obedience of the second Adam, through the renewal of the cosmos — and not the abolition of the constitution.⁶

⁵Cornelius Plantinga, *Not the Way It's Supposed to Be: A Breviary of Sin* (Grand Rapids: Eerdmans, 1995). Plantinga's recovery of the older Reformed instinct that the dissonance the world generates against our sense of its proper shape is the diagnostic signature of fallenness is the appropriate background for the present paper's treatment of common ground with the transhumanist diagnosis.

⁶The location of the etiology in *rebellion* rather than *constitution* is the consistent witness of the Reformed tradition; see Calvin's treatment in *Institutes* II.i and the dogmatic development in Bavinck, *Reformed Dogmatics*, vol. 3, on the doctrine of sin.

The depth of the disagreement is the depth of the diagnosis, and the depth of the diagnosis is the depth of the grounding. Paper 1 identified seven problems and located the first — the **grounding problem**, *what kind of thing a human is* — as the unit on which the other six load. The grounding problem is exactly the cleavage point.

In plain terms: if a human is, at root, a biological substrate that produces consciousness as the side-effect of an undirected evolutionary process, then the wrongness is in the substrate and the cure is in its replacement — and the transhumanist diagnosis follows from the materialist grounding with no remainder. If, instead, a human is an image-bearer placed under a Creator who has spoken concerning good and evil, then the wrongness is the rupture of the conferring relation and the cure is its restoration — and the Christian diagnosis follows from the Reformed grounding with the same internal coherence. Plantinga’s instinct that the world is not the way it is supposed to be does its diagnostic work precisely because it presupposes the grounding the transhumanist denies.⁷ Problem #1 is not retired here but re-engaged at the diagnostic level: it is the *first* point at which the two must disagree, and the point on which every subsequent disagreement loads.

This is the cleavage. *Same ache. Opposite groundings. Opposite etiologies. Opposite cures.* The diagnosis dictates the cure and the grounding dictates the diagnosis. If the problem is biology, the program is engineering. If the problem is rebellion, the program is reconciliation. The two cannot be combined, because they have agreed at no point past the bare observation that something is wrong. The Christian agreement with the transhumanist that humans are not what they should be must therefore be carefully bounded, lest the agreement be heard to extend to the diagnosis, which it does not, or to the cure, which it cannot.

§3. The Program at Its Candid Edge — and Its Softer Variants

This section draws out the implications the program’s logic permits, distinguishes what it permits from what its practitioners must hold, and engages the egalitarian transhumanist who explicitly denies the harshest conclusions.

Honesty requires that the transhumanist position be allowed to draw out its implications without theological softening — and, equally, that we not attribute to every transhumanist the harshest conclusions the program’s logic would license. The distinction between what a program *permits* and what its adherents *must* hold is the difference between a fair critique and a straw man, and the paper means to stay on the right side of it.

One candid edge of the program’s logic would allow the following conclusions, even if many practitioners resist them. First, that the serpent of Genesis 3 was telling the truth: the promise that eating the fruit would make the human as God, knowing good and evil, was not a deception but a delayed prophecy whose fulfillment requires only the maturing of technique. Second, that the protoevangelium of Gen 3:15 is not promise but sentimentalism — a mythic consolation offered to a species that had not

⁷Cornelius Plantinga, *Not the Way It’s Supposed to Be: A Breviary of Sin* (Grand Rapids: Eerdmans, 1995). Plantinga’s recovery of the older Reformed instinct that the dissonance the world generates against our sense of its proper shape is the diagnostic signature of fallenness is the appropriate background for the present paper’s treatment of common ground with the transhumanist diagnosis.

yet developed the means to redeem itself. Third, that the image of God is at best a primitive category and at worst an obstacle: a designation that grounds worth in something other than capability, and so impedes the rational reallocation of resources and rights to the more-capable beings the program intends to produce. Fourth, that worth itself is a function of capability and nothing else, so that the un-augmented occupy a lower rung in the moral hierarchy than the augmented.⁸ Fifth, that Babel — the post-Fall project of building a tower whose top reached into heaven and making a name for ourselves (Gen 11:1–9) — was a noble undertaking prematurely interrupted, and the present program is its rightful resumption.

These are the conclusions the program *permits* at its candid edge. The dignity-distribution problem (Problem #5 in the series' spine) bites here with unusual force: a world in which worth tracks capability is a world in which the distribution of dignity is a hierarchy, and one whose lower rungs are populated by those whom the augmented have surpassed.⁹

But fairness requires the next move, not a stopping here. **Many transhumanists explicitly reject the stratifying conclusion**, and the strongest forms of the program are at pains to preserve equal moral status. The *Transhumanist Declaration* grounds itself in universal rights and “the well-being of all sentience,” advocating a moral vision that respects “the interests and dignity of all people.”¹⁰ Nick Bostrom, in “In Defense of Posthuman Dignity,” argues directly that neither the means nor the ends of enhancement endanger human dignity, and that a properly inclusive concept of dignity extends to the enhanced and the unenhanced alike — he denies, in his own terms, that the program degrades the worth of the un-augmented.¹¹ James Hughes’s *Citizen Cyborg* develops an explicitly *democratic* transhumanism, in which enhancement must be made available to everyone, coupled with universal healthcare and an expansion of human rights, precisely so that enhancement does not stratify the species.¹² These are not fringe disclaimers; they are serious attempts to hold the

⁸This is the implication the transhumanist program permits more candidly than many of its practitioners are willing to state. The point is structural: a program whose premise is that worth tracks capability cannot, without internal contradiction, hold that the worth of the un-augmented and the augmented is equal. For the egalitarian transhumanist’s attempt to hold both, and the present paper’s coherence reply, see §3 with notes 34–36.

⁹For the dignity-distribution problem in its theological register, see Plantinga, *Not the Way It’s Supposed to Be*, and the wider literature on theological anthropology Paper 1 surveys.

¹⁰*The Transhumanist Declaration*, adopted by the Humanity+ Board (2009; originally drafted 1998 by the World Transhumanist Association). The Declaration grounds itself in universal rights and “the well-being of all sentience,” advocating policy guided by “a responsible and inclusive moral vision... showing solidarity with and concern for the interests and dignity of all people around the globe.” Cited here as a primary movement document (a manifesto, not a peer-reviewed source) evidencing that the program, in its self-understanding, intends to preserve rather than stratify dignity.

¹¹Nick Bostrom, “In Defense of Posthuman Dignity,” *Bioethics* 19, no. 3 (2005): 202–214, doi:10.1111/j.1467-8519.2005.00437.x. Bostrom argues that neither the means nor the ends of enhancement endanger human dignity, and that a properly inclusive concept of dignity extends to both enhanced posthumans and the unenhanced — i.e., he denies that transhumanism degrades the dignity of the un-augmented. The present paper engages this directly as the strongest peer-reviewed statement of the dignity-preserving variant, on grounds of internal coherence rather than motive (§3).

¹²James J. Hughes, *Citizen Cyborg: Why Democratic Societies Must Respond to the Redesigned Human of the Future* (Cambridge, MA: Westview Press, 2004). Hughes’s “democratic transhumanism” argues that enhancement must be made available to everyone, coupled with universal healthcare, a guaranteed basic income, and the expansion of human rights — an explicitly egalitarian framing premised

program and equal dignity together.

The paper's reply is therefore not *motive* but *internal coherence*. The egalitarian transhumanist wants two things at once: that worth is, at bottom, a function of capability (the premise that makes enhancement a moral imperative rather than a consumer preference), and that the worth of the un-augmented remains equal to the worth of the augmented (the egalitarian commitment). The difficulty is that the second claim has no ground in the first. If dignity is conferred — given to the creature by the Creator and not earned by the creature's powers — then the equal worth of the un-augmented is secured at its root, and the Christian can affirm the egalitarian's instinct while denying his premise. But if dignity tracks capability, the egalitarian commitment is a borrowed intuition with no home in the system that holds it: a dignity-floor imported from a tradition the program has otherwise left behind. The egalitarian transhumanist is, on this reading, more humane than his premises and is keeping faith with a conferred-dignity instinct his own grounding cannot supply. That is not an accusation of bad faith. It is the observation that the most attractive form of the program is unstable in the direction of the conferred-status anthropology it has rejected — and stable only if it borrows back what it denied.

A serious theological response must therefore engage the program at both strengths: at the strength of what its candid edge permits, *and* at the strength of what its best practitioners actually hold. The next sections do the first by exposing the structural shape of the offer; this section has done the second by showing that the program's most humane variant is humane on borrowed credit.

§4. Our Counter: The Fall, Read Through the Spine

The remainder of the paper develops the counter-position. It does so by walking Genesis 3 and a small number of adjacent texts — Gen 9:6, Gen 11:1-9, Rom 5:12-14, Rom 8:19-22 — and showing that the seven-problem set Paper 1 grounded in creation is *confirmed*, not abolished, by the Fall. The Fall does not retire the problems; it intensifies and universalizes them. It also distorts them in a particular direction — the direction the transhumanist program now follows to its terminus. Reading the Fall through the spine therefore does two things at once: it answers transhumanism by exposing the structural shape of its offer, and it sets up the recovery the rest of the series will trace.

4a. The Serpent's Offer: *Eritis Sicut Deus* (Gen 3:1-7)

Here is how Problem #2 (displacement) looks under the Fall: the route to godlikeness is inverted — grasped by autonomous action rather than received in conferred relation.

"You will not surely die. For God knows that when you eat of it your eyes will be opened, and you will be like God, knowing good and evil" (Gen 3:4-5, ESV). The serpent's offer is the origin point of every subsequent program of self-deification, and its structure repays attention. The lie is not that godlikeness is bad. The first chapter of Genesis has already located humanity in the image and likeness of God;

on preserving, not stratifying, equal status.

an aspiration to be *like* God is, in one register, the proper aspiration of the image-bearer. The lie is the structural inversion: that godlikeness is a thing to be grasped by autonomous action rather than received in conferred relation. The serpent does not deny the human telos. He counterfeits the route to it.

In plain terms: the problem is not wanting to be like God; it is trying to get there by seizing power instead of receiving it.

This is precisely the structural shape of the modern transhumanist offer. Transhumanism does not deny that humans aspire to transcend their limits; it affirms that aspiration and proposes a route. The route, in both cases, is the same: grasp by capability what was meant to be received by relation. The image-bearer was made to mature into the fullness of the image *under God*; the serpent proposes — and the transhumanist refines — a maturation that bypasses the conferring relation altogether. This is what Paper 1 called Problem #2, displacement, at its taproot.

A precise note about the two papers' relation is owed here. Paper 1 dissolved Problem #2 as an *independent syllogism*: the inference from "X is more capable than the human" to "X displaces the human" has no force once grounding is understood as conferred rather than achieved. The present paper does not re-open that syllogism. It engages #2 in a different register — as a Fall-historical *impulse* whose genealogy the canonical narrative discloses. The logical death of the displacement inference and the Fall-historical persistence of the displacement impulse are not in tension; they are the two registers in which the same problem must be addressed. Every subsequent displacement of conferred status by capability runs in the channel the serpent cut.

The exegetical point should not be hurried past. The fruit is not, in the narrative, a magical accelerator of biological development. It is the symbol of a human refusal to live under the word of God concerning good and evil. The eating is an act of epistemic and moral autonomy: the human will now know good and evil for himself, by his own determination, and not as a creature receiving these categories from his Maker. The transhumanist program, taken at full strength, is the institutional and technological elaboration of exactly this refusal — structurally not a new offer but the old one, well-funded; the same epistemic move undertaken with very different instruments.

Three features of the serpent's offer repay further attention, because each surfaces in the modern register.

First, the offer trades on a partial truth. The serpent does not say "you will not die" simply; he says "you will not surely die," and "your eyes will be opened, and you will be like God" — and the opening of the eyes (Gen 3:7) does in fact take place. Something is delivered. The human after the eating is in possession of a knowledge he did not have before. The serpent's reliability as a source of information is therefore not zero, and this is precisely what makes the offer dangerous: it is not a simple lie refutable by checking the outcome, but a structural distortion that delivers, in part, what it promises while concealing the cost. The transhumanist program has the same structure. It will deliver, in part, what it promises: lifespans will lengthen, capacities will expand, suffering will in measurable senses decrease. The Christian response cannot be the false prediction that nothing will be delivered. The Christian response is the careful naming of *what is delivered* and *what is paid for it* — and the cost, on the present argument, is the constitution of the human as image-bearer under God,

exchanged for the constitution of the human as self-determining augmentee. The serpent's offer was true in its surface and false in its deep grammar. The modern offer has the same shape.

Second, the offer is received, not merely heard. The text registers Eve's deliberative process: "So when the woman saw that the tree was good for food, and that it was a delight to the eyes, and that the tree was to be desired to make one wise, she took of its fruit and ate" (Gen 3:6, ESV). The seriousness of the temptation lies in the genuineness of the goods it offers — food, beauty, wisdom — and in the orientation of the human faculties toward those goods, an orientation that before the offer was properly aimed at them under God. The Fall is not the choice of evil over good; it is the choice of good *outside* the conferring relation rather than *within* it. The transhumanist offer succeeds, where it succeeds, by appealing to the same faculties oriented toward genuine goods — health, longevity, expanded knowledge — apart from the relation under which alone those goods are properly received. A theology of the Fall that locates the offense in the desire for the goods rather than in the inversion of the relation will misdescribe the structure of every subsequent temptation.

Third, the offer is mediated by a creature speaking to creatures about the Creator. The serpent is not God; the serpent is a creature within the order God has made, who proposes to the human a reading of God's word ("Did God actually say...?") and then a reading of God's motives ("for God knows..."). The Fall is, in part, a hermeneutical failure: the human accepts a creaturely interpretation of the divine word as adequate ground for action against the divine word itself. This is structurally important for the modern register, where the offer comes again from creatures — from very capable ones, impressively credentialed, whose competence in their own domains is real — about what the human is for and what the human ought to become. The credentialed technologist who has acquired genuine competence in molecular biology, in computer science, in the neural correlates of consciousness, has thereby acquired no authority over the anthropological question of what the human is to *become*. Competence within a domain is not authority over the constitution of the creature; the conflation of the two is the Fall's epistemic move in modern dress. This is not anti-intellectualism. It is the recognition that *what is the human to become* is not the kind of question competence within the human order is positioned to answer. The serpent's structural feature — *creature interpreting Creator to creatures* — is the structural feature of every later iteration of the offer, and the modern iteration is the most credentialed it has ever been.

4b. The Image Defaced but Not Erased (Gen 9:6 and the *Imago* Tradition)

Here is how Problem #3 (the unit of analysis) survives the Fall: the image is marred, not lost; it remains the floor under every human, augmented or not.

The Fall does not abolish the image of God in humanity. This is the explicit teaching of Gen 9:6, spoken not before but *after* the cascade has worked itself out through Cain, through the antediluvian violence, and through the flood: "Whoever sheds the blood of man, by man shall his blood be shed, for God made man in his own image" (ESV). The prohibition on murder, given to a humanity whose fallenness has by this point been amply demonstrated, is grounded in the persistence of the *imago*. Whatever

the Fall has done to the image, it has not erased it.¹³

The tradition's instrument for holding this together is the distinction between the image as *structure* and the image as *conformity*. The image is not a possession the human carries alongside others; it is what the human *is*, and it cannot be lost without the human ceasing to be human, which the texts do not permit.¹⁴ What is lost — or rather marred — is the conformity of the human to the image: the actual likeness in righteousness, holiness, and true knowledge that the human was made for and which, in the Fall, has been distorted. The image as structure remains; the image as conformity is broken.

This pair is not bookkeeping; it is the conceptual instrument by which the tradition holds two truths together: that the human is, after the Fall, still the kind of creature he was made to be (image as structure), and that the human is no longer in fact what he was made to be (image as conformity, distorted). The long patristic and medieval working-out of the distinction is the historical ground on which the Reformed instinct was clarified.¹⁵

What is at stake, on the present argument, is the answer to the deepest version of the transhumanist claim. The transhumanist does not, in his most candid form, claim merely to *improve* the human; he claims to *transcend* the limits constitutive of the human as he now stands. The answer is that the image as structure *is* the human and is not a feature merely carried by the human. A being who has abandoned that constitution has not transcended humanity but has departed it. Hoekema's "perversion but not destruction" formulation and Bavinck's insistence that the image is what the human *is* converge on this: the project of transcending the constitution is not the perfection of the image but its abandonment, and the augmentee at the project's terminus is not a higher human but a different kind of thing.

In plain terms: the candid transhumanist's own framing — that the goal is to become posthuman — concedes the point. The choice the program permits, at its edge, is not "stay human or become more human"; it is "stay human or cease to be one."

For the present paper, this matters because it is the load-bearing answer to two opposite errors. The first — the transhumanist's, that the image is a primitive category to be transcended — has just been addressed. The second, opposite error is the despairing reader's: that the Fall has left the human without dignity or worth. To this the answer is again Gen 9:6, where God himself grounds the prohibition on murder, after the Fall, in the persistence of the image. The image is the floor under every human regardless of capability, and it is the unit of analysis — Paper 1's Problem #3 — that the cascade has marred but not removed. When the *imago* anchors political and legal anthropology — as in Grotius's deployment of it as the ground for natural-law

¹³Anthony A. Hoekema, *Created in God's Image* (Grand Rapids: Eerdmans, 1986). Hoekema's "perversion but not destruction" formulation is the load-bearing one for the present paper.

¹⁴Herman Bavinck, *Reformed Dogmatics*, vol. 3: *Sin and Salvation in Christ*, ed. John Bolt, trans. John Vriend (Grand Rapids: Baker Academic, 2006), on the doctrine of the *imago* and its persistence through the Fall.

¹⁵Isabella Image, *"The Human as Imago Dei"* (Oxford University Press, 2017), tracing the patristic and medieval reception of the *imago* within which the Reformed structure-conformity distinction was clarified. Image's study is the appropriate historical-reception ground for the present argument that abandonment of the image-as-structure is departure from humanity rather than transcendence of it.

claims about the inviolability of persons — the worth of the un-augmented and the augmented is structurally identical.¹⁶

4c. Vocation Cursed: Work Curves Toward Self-Glory (Gen 3:17-19; Gen 11:1-9)

Here is how Problem #4 (vocation) looks under the Fall: work remains, but its telos is bent toward the name of the builders.

The fourth problem of the spine — vocation, the work the human is for — is not abolished by the Fall but cursed and distorted. “Cursed is the ground because of you; in pain you shall eat of it all the days of your life; thorns and thistles it shall bring forth for you” (Gen 3:17-18, ESV). Work remains; the relation of the worker to the ground he works has changed. The dominion commission of Gen 1:28 is not revoked; it is rendered laborious, frustrated, and entropic in its execution. The cursed vocation does not lie dormant: it is taken up by the human under a new orientation — the worship of the self in place of the Creator — and the work that results bears the shape of that worship.¹⁷

Genesis 11 narrates the result with precision. “Come, let us build ourselves a city and a tower with its top in the heavens, and let us make a name for ourselves, lest we be dispersed over the face of the whole earth” (Gen 11:4, ESV). The post-Fall vocation of humanity is here organized — collectively, technologically, and impressively — for a purpose the chapter labels with three telling phrases. The construction is *for ourselves*; the name is *for ourselves*; the dispersal feared is the dispersal that would interrupt the project. Vocation has been inverted at the level of telos. The image-bearer was made to exercise dominion under God; the Babel-builder exercises dominion for the name of the builders. The religious instinct, in fallen humanity, does not vanish but bends, producing idols rather than worship — and the vocational instinct bends with it.¹⁸ The work-making faculty does not vanish in the Fall. It is bent.

A careful word is required here against anachronism. **Babel is not literally transhumanism.** The eleventh chapter of Genesis is not a coded prediction of a technology not yet invented, and the paper does not argue that it is. The argument is structural. Babel is a *Babelic* moment in human history, and the transhumanist program is structurally Babelic: a collective, technologically ambitious, self-glorifying construction undertaken in the register of *eritis sicut Deus*. The structural identification is exegetically careful precisely because it does not collapse the distance between the texts and the present. The Bible’s eschatological geography is shaped by the contest between the city-the-builders-build and the city-God-builds, and the post-Fall vocation

¹⁶Janne E. Nijman, “Grotius’ Imago Dei Anthropology” (Oxford University Press, 2017), traces the function of the *imago* in Hugo Grotius’s natural-law construction. The point relevant here is that when the *imago* anchors political anthropology, the worth of augmented and un-augmented humans is structurally identical.

¹⁷G. K. Beale, *We Become What We Worship: A Biblical Theology of Idolatry* (Downers Grove: IVP Academic, 2008). Beale’s exegetical-biblical-theological case that the human is conformed to the object of his worship is the load-bearing background for the present argument that the cursed vocation is taken up under a new orientation.

¹⁸John Calvin, *Institutes of the Christian Religion*, II.i-ii, on the bent state of the *sensus divinitatis* in fallen humanity.

of humanity stands under that contest from Gen 11 forward.¹⁹ The transhumanist program belongs to the lineage of the builders' city. Naming it as one *instance* of a recurring structural pattern is not anachronism; treating it as the prophesied fulfillment of Genesis 11 would be. The paper claims the former and disavows the latter.

Paper 1's fourth problem — vocation — is therefore featured here in its post-Fall register. The human still has work; the work has been cursed; the curse has not abolished the vocation but has bent its telos. The recovery of vocation under God is the work of subsequent papers and is foreshadowed but not developed here.

4d. Governance Distorted: Dominion, Domination, and the Mirror Sin of Abdication (Gen 3:16)

Here is how Problem #6 (governance) looks under the Fall — and this is the section that bears most directly on autonomous systems. Fallen governance bends two ways: over-reach (claiming office on the strength of capability) and abdication (surrendering office on the strength of someone else's capability). Both confuse capability with conferred office.

"Your desire shall be for your husband, and he shall rule over you" (Gen 3:16, ESV). The post-Fall statement of governance in the marriage relation is, on the standard Reformed reading, *descriptive* of the Fall's distortion of authority, not *prescriptive* of an arrangement God has ordained. The statement sits within a *judgment* speech — God speaking to the serpent, then the woman, then the man, each addressed under what the Fall has now done. The speech is not a constitutional reordering of the offices God has made; it is a description of how those offices, having been corrupted, will in fact now run.²⁰ What is fallen is not merely each human will in isolation but the *relations* through which the offices are exercised, and the bent relation produces the bent exercise.²¹ To read the verse as prescriptive — as if God were instituting husbandly domination at Gen 3:16 — is to confuse a *description of how the broken thing now works* with a *prescription for how it ought to work*, a confusion that has done historical damage out of all proportion to the size of the exegetical mistake. Dominion under God degrades to power-over-other. The governance commission — Paper 1's Problem #6 — does not disappear; it is distorted.

It is essential to name both Fall-shaped governance failures, not one.

The over-reach failure is the obvious one: domination, the use of office for the self rather than the other; the rule of the capable over the less capable on the strength of the capability itself. The transhumanist program, at full strength, has a recognizable governance shape, and it is the shape of over-reach: the augmented will govern

¹⁹Meredith G. Kline, *Kingdom Prologue: Genesis Foundations for a Covenantal Worldview* (Eugene: Wipf & Stock, 2006); T. Desmond Alexander, *From Eden to the New Jerusalem: An Introduction to Biblical Theology* (Grand Rapids: Kregel, 2008). Both develop the eschatological-geographic contest between the city the builders build and the city God builds.

²⁰Louis Berkhof, *Systematic Theology*, on the noetic and volitional effects of the Fall; the descriptive-not-prescriptive reading of Gen 3:16 follows the standard Reformed treatment of the judgment-speech of Gen 3:14-19.

²¹G. C. Berkouwer, *Sin*, trans. Philip C. Holtrop (Grand Rapids: Eerdmans, 1971), on the structural — not merely individual-moral — register of the Fall's effects on the relations through which the offices are exercised.

the un-augmented because they will be more capable, and capability is — on the program’s premises — the relevant qualification. The bent office does not merely *fail* to exercise its commission rightly; it actively *reorients* itself toward the self.²² Over-reach is the failure the present moment is least likely to *misdiagnose*, because the surrounding culture already has a vocabulary — *power, hierarchy, coercion* — for naming it. The transhumanist program’s distinctive contribution is the simplification of its rationale: earlier rationales for the rule of one over another had to gesture, however hypocritically, at something other than capability — divine sanction, consent, custom. The program proposes to drop the gesture. If worth tracks capability, so does the qualification for office. The candor is admirable; the structural identification is direct.

The mirror failure must also be named, because it is the one the present moment is more likely to commit. **Abdication** is the cession of the governing commission to a more-capable other on the same logic — that capability is the relevant qualification for office. Where over-reach claims office on the strength of capability, abdication surrenders office on the strength of someone else’s capability. Both confuse capability with conferred office. The image-bearer’s governance commission was conferred by God in the act of creation; it is not transferable to whatever being later turns out to be more capable. The prospective cession of the human dominion to autonomous systems on the grounds of those systems’ superior capacity is abdication in technological register — a Fall-shaped move whether or not the one making the cession recognizes it as such.

The abdication argument deserves further weight, because the secular disciplines that study delegated authority have, on their own terms, surfaced the same structural fact. Three converging registers — economic principal-agent theory, control-theoretic models of delegated authority, and the contemporary literature on aligning autonomous artificial systems — independently identify a single point: the delegation of action does not transfer responsibility; the agent acts on its own objective function whether or not it has been correctly aligned with the principal’s; and the principal’s authority is not extinguished by delegation but translated into a harder problem — supervising an agent whose competence in the delegated task may exceed the principal’s own.

A one-sentence gloss for each, since the terms are technical:

- *Principal-agent theory* (economics): when a principal delegates a decision to a better-informed agent he cannot fully monitor, the agent’s incentives and the principal’s can diverge; the principal’s job is not eliminated by the delegation but reshaped into incentive design and monitoring.²³

²²Bavinck, *Reformed Dogmatics*, vol. 3, on authority and its distortion, where the bent office is read not as a merely accidental failure of an otherwise sound exercise but as a re-orientation of office toward the self.

²³Michael C. Jensen and William H. Meckling, “Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure,” *Journal of Financial Economics* 3, no. 4 (1976): 305–360, is the foundational treatment of the principal-agent problem in the modern information-economics tradition, with the subsequent literature on incentive design, monitoring, and moral hazard developing the structural point at greater length. The present argument uses the framework at the conceptual level: that delegation does not extinguish responsibility but reshapes it.

- *The specification problem* (control theory / cybernetics): a control system that hands its objective to a more capable subsystem must specify that objective with a precision the higher capability will exploit; whatever is left underspecified will be optimized against.²⁴
- *The alignment problem* (AI safety): the more capable the delegated agent, the more consequential the smallest gap between the agent’s objective and the principal’s actual end.²⁵

The convergence is striking, and it serves the theological argument rather than competing with it. Three independent secular disciplines, none theologically motivated, have come to articulate something the Reformed tradition has held under the doctrine of vocation all along: the office is not transferable. The economist names it the principal-agent problem; the control theorist names it the specification problem; the alignment researcher names it the alignment problem; the theologian names it the irreducibility of conferred office. The relation between the registers must be kept clean: the secular disciplines surface the *structural fact* that delegation does not extinguish authority or responsibility; the theologian *names its ground* in the conferring relation the secular disciplines, on their own terms, cannot reach.

This does not settle the practical questions about what may and may not be delegated. The Reformed tradition has always permitted — and commended — the delegation of *tasks* under the abiding office. Jethro’s counsel to Moses (Ex 18) is the canonical case; the deacons of Acts 6 are the New Testament analog. What the tradition has not permitted is the delegation of *office* — the cession of the conferring relation itself — to any creature. The distinction is between delegating *work to be done under the office* and ceding *the office under which the work is done*. The transhumanist program, at its candid edge, contemplates the second. The cultural drift toward unsupervised algorithmic governance contemplates the second by negligence rather than intention. Both are Fall-shaped in the same way. The careful Christian response is not Luddism — the refusal of the capable instrument — but the insistence that the office remain located where the Creator located it: with the human image-bearer who answers to the Creator for its exercise.

These are not abstractions. The drift toward abdication is already documented in the systems we have built, and three near-term cases make the dignity-distribution stakes concrete.

Algorithmic lending. A peer-reviewed study of consumer-lending discrimination in the FinTech era found that algorithmic and face-to-face lenders alike charged risk-equivalent Black and Latinx borrowers higher interest rates, costing minority borrowers on the order of hundreds of millions of dollars a year — automated systems re-

²⁴Norbert Wiener, *Cybernetics: Or Control and Communication in the Animal and the Machine* (Cambridge, MA: MIT Press, 1948), as the founding work in the field whose later development surfaces the specification problem the present argument invokes. The structural point — that an underspecified objective handed to a more capable subsystem will be optimized against — is a stable result across the subsequent control-theoretic and operations-research literatures.

²⁵Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané, “Concrete Problems in AI Safety,” arXiv:1606.06565 (2016). The point relevant here is structural rather than technical: the more capable the delegated agent, the more consequential the smallest misspecification of its objective, and the more demanding the supervision the principal must mount to keep the delegated work aligned with the end the office was conferred to serve.

producing, and in some respects laundering, a stratification of treatment by group.²⁶ The office of fair judgment was, in effect, ceded to a model whose objective function was never the office's actual end.

Autonomous weapons. A United Nations Panel of Experts reported that retreating forces in Libya were “hunted down and remotely engaged” by a loitering munition “programmed to attack targets without requiring data connectivity between the operator and the munition” — among the first reported uses of a weapon in fire-and-forget mode against human targets.²⁷ The International Committee of the Red Cross has since called for legally binding rules prohibiting unpredictable autonomous weapons and those that target humans, precisely to preserve human control and human judgment over the use of force.²⁸ The contested point — whether full autonomy was exercised — is exactly the point: the abdication is the *cession of the kill decision* to the system, and the office of judgment over life and death is not one Scripture permits a creature to delegate to an instrument.

Automated welfare adjudication. Australia's Royal Commission into the Robodebt Scheme found an automated income-averaging debt-recovery program “neither fair nor legal” and unlawful under social-security legislation, with officials referred for prosecution.²⁹ The Dutch childcare-benefits scandal saw a risk-scoring algorithm use nationality as a fraud-risk factor, wrongly accusing tens of thousands of families.³⁰ Michigan's MiDAS system issued tens of thousands of false fraud accusations at an error rate later documented above ninety percent.³¹ In each case the office of just adjudication over the vulnerable was abdicated to a system, and the human authority

²⁶Robert Bartlett, Adair Morse, Richard Stanton, and Nancy Wallace, “Consumer-Lending Discrimination in the FinTech Era,” *Journal of Financial Economics* 143, no. 1 (2022): 30–56, doi:10.1016/j.jfineco.2021.05.047. The study documents that both algorithmic and face-to-face lenders charged risk-equivalent Latinx and Black borrowers higher interest rates, with an estimated cost to minority borrowers on the order of hundreds of millions of dollars annually.

²⁷United Nations Security Council, *Final Report of the Panel of Experts on Libya Established Pursuant to Security Council Resolution 1973 (2011)*, S/2021/229 (8 March 2021). The report describes retreating logistics convoys and forces being “hunted down and remotely engaged” by loitering munitions, including the STM Kargu-2, “programmed to attack targets without requiring data connectivity between the operator and the munition.” The degree of autonomy actually exercised is contested; the paper cites the case for the structural point about ceding the engagement decision, not to assert a confirmed autonomous killing.

²⁸International Committee of the Red Cross, “ICRC Position on Autonomous Weapon Systems” (12 May 2021). The ICRC recommends new legally binding rules: prohibiting unpredictable autonomous weapon systems and those designed or used to apply force against persons, and restricting others, in order to preserve meaningful human control and human judgment over the use of force.

²⁹Royal Commission into the Robodebt Scheme, *Report* (Commonwealth of Australia, 7 July 2023), Commissioner Catherine Holmes AC SC. The Commission found the automated income-averaging debt-recovery scheme to be neither fair nor legal and unlawful under the relevant social-security legislation, and made referrals for civil and criminal prosecution.

³⁰Amnesty International, *Xenophobic Machines: Discrimination Through Unregulated Use of Algorithms in the Dutch Childcare Benefits Scandal* (London: Amnesty International, 2021). The Dutch tax authority's risk-scoring system used non-Dutch nationality as a fraud-risk indicator, contributing to the wrongful accusation of tens of thousands of families (the *toeslagenaffaire*).

³¹On Michigan's MiDAS unemployment-fraud system, which issued tens of thousands of false fraud determinations at an error rate documented above ninety percent, see the litigation in *Cahoo v. SAS Analytics Inc.* and *Bauserman v. Unemployment Insurance Agency* (settled 2022), and the scholarly treatment in “Automated Stategraft: Faulty Programming and Improper Collections in Michigan's Unemployment Insurance Program,” *Wisconsin Law Review*.

that should have answered for the judgment was diffused into the machinery until no one answered for it at all.

The point is not that the Christian must claim power against the capable; it is that the Christian must distinguish capability from office. Paper 1's sixth problem — governance — is featured here in its distorted form: by over-reach in the form transhumanism makes available, and by abdication in the form the surrounding culture is increasingly tempted to perform. Both are answered, ultimately, only by the recovery of conferred office under the second Adam, which is the work of subsequent papers.

4e. Federal Headship: The Cascade From One Man (Rom 5:12-14)

Here is how Problem #1 (grounding) is extended into covenantal register: the universality of the cascade is not contingent but federal — traceable to the one in whom the race stands.

The Fall is not Adam's private tragedy. "Therefore, just as sin came into the world through one man, and death through sin, and so death spread to all men because all sinned" (Rom 5:12, ESV). Paul's compressed argument in Rom 5:12-14 is that the entry of sin into the human race is inseparable from the entry of death, and that both are traceable to the act of *one man*, in whose act the race stands as in a federal head. The grammar of v. 12 has been long discussed, but the structural point is not in dispute among those who read it within its own Pauline frame: the consequence of Adam's act extends beyond Adam, and it does so by a relation properly described as covenantal and imputational rather than merely exemplary.³² Paul is working with a typology in which the *one* stands for the *many* — first negatively in Adam, then, by the same logic, positively in Christ.³³ The dogmatic side of the same point has been developed with the precision the question requires,³⁴ and with the further service of distinguishing what the doctrine must affirm from what it has sometimes been overburdened with, while preserving the federal structure as load-bearing.³⁵

A binding scope note is necessary. Paul's argument continues, in vv. 15-21, into the "much more" of the second Adam, where the recovery is developed — the obedience of the one in whom the *many* are made righteous, as in the disobedience of the one the *many* were made sinners. That recovery is the load-bearing argument of the next paper and is not developed here. The present paper restricts itself to vv. 12-14 — the cascade *from* Adam, downward into the race. The recovery *through* Christ is foreshadowed, and the typology named, so that the paper's argument terminates at the correct boundary and does not pretend the cascade is the last word. It is not the last word. But it is the *present* word, and the careful exegete will distinguish what the Fall has done from what the second Adam has accomplished, lest the comfort of the latter obscure the diagnosis of the former.

³²The grammatical question concerning the antecedent of *eph hō* in Rom 5:12 is the focus of the discussion; the structural point — that the consequence of Adam's act extends covenantally — is held across the major Reformed readings independent of the grammatical decision.

³³James D. G. Dunn, "Adam and Christ," in *Reading Paul's Letter to the Romans*, ed. Jerry L. Sumney (Atlanta: SBL Press), 125-138, doi:10.2307/j.ctt32bzb7.12.

³⁴John Murray, *The Imputation of Adam's Sin* (Phillipsburg: Presbyterian and Reformed, 1959).

³⁵Henri Blocher, *Original Sin: Illuminating the Riddle* (Downers Grove: IVP, 1997).

Why does this matter for the answer to transhumanism? Because the universal scope of the seven-problem set — which the transhumanist program meets at every point — is not a cultural or contingent universality. It is covenantal. The displacement of conferred status by capability, the distortion of vocation toward self-glory, the bending of governance toward domination or abdication: these are not the local failings of one civilization that another, more enlightened, might engineer its way past. They are the shape of fallen human existence, traceable to the head in whom the race stands, and they are not addressable by any program that does not deal with the head. The transhumanist program proposes to deal with the symptoms by upgrading the substrate. The Christian gospel proposes to deal with the head by furnishing a new one. The two are not in competition at the level of strategy; they are in competition at the level of what the problem is.³⁶

4f. Cosmic Entropy: Creation Subjected to Futility (Rom 8:19-22)

Here the cascade is shown to be cosmic, not merely human — and the section is careful about what it does and does not claim about physical decay.

The cascade is not anthropological only. It is cosmic. “For the creation was subjected to futility, not willingly, but because of him who subjected it, in hope that the creation itself will be set free from its bondage to corruption and obtain the freedom of the glory of the children of God. For we know that the whole creation has been groaning together in the pains of childbirth until now” (Rom 8:19-22, ESV). The Fall ripples beyond the human and into the material order. The non-human creation has been subjected — not by its own act, but by God in response to the human act — to a *futility* whose hopeful reverse-orientation toward the eschatological liberty of God’s children is the only reason the subjection is not despair.

Three exegetical features do not survive being passed over.

*First, the verb *hypetagē* — “was subjected” — is a divine passive.* The creation did not, of its own act, become futile; it was *subjected* to futility, and Paul specifies that the subjection was *not willing* and that the subjecter is “him who subjected it.” The subjection is a *judgment* — God’s response to the human act of Gen 3 — and the agency is God’s, not the creation’s, nor the serpent’s, nor any abstract entropic principle’s.³⁷ This locates the cascade’s origin where the canonical narrative does: not in a defect of the creation’s design, but in a divine response to human rebellion.

Second, the subjection is in hope. The phrase *eph’ helpidi* is not decorative. The same verse that names the subjection names also its forward orientation: the creation was subjected with a view to a future liberation, and the subjection itself carries that orientation as part of its structure. The futility of the present cosmos is not its last word, and the eschatological liberty of the children of God is not an afterthought added to the futility but the orientation in which the futility was imposed.³⁸ This is the struc-

³⁶Douglas J. Moo, *The Epistle to the Romans*, New International Commentary on the New Testament (Grand Rapids: Eerdmans, 1996), *ad loc.* Rom 5:12-14. Moo on Rom 5:15-21 is reserved for Paper 3.

³⁷Moo, *Romans*, NICNT, *ad loc.* Rom 8:19-22.

³⁸David Wilkinson, *Christian Eschatology and the Physical Universe* (London: T&T Clark, 2010), doi:10.5040/9780567660497. The point relevant here is that the futility of Rom 8 is not a rewriting of natural law but a subjection of the created order to a corruption whose final reversal is the resurrec-

tural hinge into §7: the cascade is entropic, and the cascade is also, by the same divine speech that subjected it, oriented toward its own reversal. The protoevangelium of Gen 3:15, spoken inside the judgment of Gen 3, is the anthropological face of the same orientation Rom 8 names cosmically. To read the futility without the *in hope* is to misread it; to read the *in hope* without the futility is to evade the cost.

Third, the groaning is the groaning of childbirth. The metaphor is productive, not merely degenerative. The pains the apostle names are the pains of a creation in labor — the cost of what is being brought forth, not the symptoms of a system in terminal failure.³⁹ This matters for the transhumanist engagement in particular. The transhumanist reads the groaning as the lived face of a defect to be engineered away; the apostle reads it as the lived face of a creation in productive travail under God. The two readings cannot be combined, because they differ at the level of what the groaning is *for*.

A necessary clarification about physical decay, since it is the easiest point to misread. The paper does not claim that the second law of thermodynamics began at the Fall, nor does it adjudicate whether physical decay obtained in the world before Genesis 3. Scripture is not committed to any particular microphysical account of entropy. What the doctrine of the cascade claims is *teleological and covenantal*, not microphysical: that the order now bears, under judgment, the status of *futility* — that decay and death now stand as judgment-with-hope rather than as the neutral running of an indifferent system. The distinction is worth stating plainly. “The physical second law describes how this present order in fact runs” is a claim of natural science. “God subjected this order to futility in response to sin, in hope of its liberation” is a theological claim about the order’s teleological status before its Maker. The two operate at different levels and do not compete; the next section develops their relation, and is careful not to collapse them.

The title of this paper — *The Entropic Cascade* — is owed to this passage. And here is the point that returns the section to the opening engagement with transhumanism: limitation and mortality are *symptoms* of the cascade. They are not the disease. Engineering symptoms while leaving the disease untreated is, at its most generous, a misdiagnosis; at its most candid, the serpent’s logic at industrial scale. The transhumanist program offers to lift the futility from the creature by lifting the creature out of the creation. The Christian gospel offers to lift the futility from the creation by reconciling the creation to its Creator. The two are not competing strategies for the same end. They are competing ends.

4g. Entropy and Its Scientific Registers: Physics, Biology, Information

This section argues that the futility Rom 8 names has recognizable analogues in three sciences — and it is deliberately modest about the strength of the claim.

The title of this paper would be merely literary if the futility Rom 8 names had no recognizable analogue in the sciences that describe the created order under their own

tion.

³⁹Klaus Nürnberger, “The ‘Spirit’ and the ‘Groaning of Creation’ in Romans 8,” on the Spirit’s role in the groaning of creation in Rom 8 and the pneumatological pledge of the future liberation.

categories. It has. The brief survey that follows is not natural philosophy for its own sake but a structural observation: that *mataiotēs* — the futility to which creation has been subjected — can be read as the theological name for a feature of the created order that physics, biology, and information theory have, each in its own register, independently noticed. The relation is one of *illuminating analogy*, not identity, and the section claims no more than that. Scripture is not borrowing a scientific metaphor, and the sciences are not secretly doing theology; the suggestion is that the same created order, examined under different lights, shows a recognizably similar structural feature.⁴⁰

Physics. The second law of thermodynamics — in the form Clausius stated it in the mid-nineteenth century and Boltzmann reframed statistically at its close — holds that the entropy of an isolated system tends to increase. Useful energy degrades into less useful forms; ordered configurations tend, overwhelmingly, toward less ordered ones; the arrow of time, as the working physicist understands it, is the asymmetry the second law establishes. The cosmological extrapolation is the so-called heat death: the asymptotic condition of a universe in which no further work is extractable from any gradient. The picture is one of irreversible degradation built into the running of the closed system. The analogy with Rom 8 is *suggestive at the structural level*: the apostle says the creation was *subjected* to futility — the passive verb is load-bearing — and that the subjection is *in hope* of a liberation that does not arise from within the subjected order. The thermodynamic picture, on its own terms, has no internal resource for hope; the order it describes runs to its terminus without remedy. The theological picture agrees about the subjection but locates the hope in what is given to the order from beyond it — the resurrection that introduces a new economy. The two pictures need not be in tension. The careful claim is not that Rom 8 *is about* thermodynamics, nor that the second law is the *mechanism* of the curse, but that the order Scripture calls subject-to-futility is the same order the physicist describes as running down, read under a different light.

Biology. Schrödinger's 1944 *What Is Life?* offered the now-classic formulation that a living organism maintains itself far from thermodynamic equilibrium for a finite period by feeding on what he called *negative entropy* — importing order from its environment and exporting waste at higher entropy than it entered. Life, on this account, is local resistance to the universal thermodynamic trend, sustained for a time and finally lost; death is the moment the negentropic resistance fails and the second law re-asserts itself locally; aging is the cumulative degradation of the systems by which

⁴⁰Wilkinson, *Christian Eschatology and the Physical Universe*. The convergence-of-registers argument the present section develops is in the spirit of the monograph's project, which holds that the natural-scientific picture of the cosmos and the eschatological promises of the New Testament illuminate one another rather than standing in tension. The classic statements of the thermodynamic argument referenced in the body are Rudolf Clausius's mid-nineteenth-century formulations of the second law and Ludwig Boltzmann's late-nineteenth-century statistical reframing. Erwin Schrödinger, *What Is Life? The Physical Aspect of the Living Cell* (Cambridge: Cambridge University Press, 1944), is the canonical statement of the negentropy argument for living systems. Claude E. Shannon, "A Mathematical Theory of Communication," *Bell System Technical Journal* 27 (1948): 379–423, 623–656, is the originating statement of the information-theoretic entropy measure. The relation between thermodynamic and information entropies, surfaced in Rolf Landauer's work on the thermodynamic cost of information erasure, is treated in the standard literature on the foundations of computing and physics. As the body states, the section claims analogy, not identity.

the resistance is mounted. None of this is in dispute as a description. The theological reading does not contest the description; it contests the interpretation. The transhumanist reads the negentropic resistance as a project to be extended indefinitely — pushing the moment of failure past the visible horizon. The Christian reading differs at the root: the resistance is finite *by design*, the design is not a defect, and the transformation that overcomes death is not the indefinite extension of the resistance but the introduction of a body, in resurrection, whose relation to entropy is not the present body's. The transhumanist offers more time within the cascade. The Christian gospel offers a body outside it. These are different outcomes, not competing strategies for the same one.

Information. Shannon's 1948 *Mathematical Theory of Communication* introduced an entropy measure — $H = -\sum p \log p$ — quantifying the uncertainty in a probability distribution, or equivalently the information content of a signal source. The mathematical and physical entropies are not the same quantity, though the literature on Landauer's principle shows they meet at the level of physical implementation: erasing information has a thermodynamic cost. For the present argument the relevant point is structural, and it must be held lightly. Communication channels degrade; signals are corrupted by noise; the shared codes by which messages are decoded across populations fragment. The post-Fall narrative of Gen 11:1-9 records, as a judgment on Babel, the multiplication of tongues — the moment the human linguistic order is fragmented into incomprehensibility. This *can fruitfully be pictured* — as heuristic analogy for the modern reader, not as a claim that the text is describing an entropy event in the technical sense — as a collapse of channel capacity and a loss of the shared code between groups: the mutual information that made cross-group decoding possible is destroyed, signals once carrying meaning become noise to those outside the speaker's basin, and coordination at scale becomes intractable. The text does not use the modern category, and the paper does not pretend it does. What the picture surfaces is only this: the cascade reaches into the order of human communication as well, and the same structural shape — the degradation of *usable* order — appears in the linguistic register as in the thermodynamic and biological ones.

The convergence — stated modestly. Three independent scientific registers, developed without theological motivation, can be read as noticing a single feature: the created order, as it now stands, exhibits a structural tendency toward the degradation of usable order. Thermodynamic systems run toward heat death; biological systems hold the trend off locally and finitely and then succumb; communicative systems are subject to noise and to the fragmentation of shared codes. The doctrine of the cascade does not *require* the sciences in order to be true, and the paper does not rest its theology on them. But the sciences, taken on their own terms, supply an unsought analogue to what the doctrine has named all along: that the order under which we live is one in which futility is, in some real sense, built into its present operation, and that the human experience of decay, mortality, and the failure of our most ambitious coordinations is not a contingent misfortune but the lived face of a structural condition.

The convergence has two consequences for the argument. First, it reframes the transhumanist diagnosis from the inside. The transhumanist is correct that the present order is one in which limitation, decay, and death are universal and structural. The

transhumanist is incorrect to infer that the appropriate response is the engineering of an exemption from the structure. The structure is not, on the Christian reading, the disease; it is the present face of a creation under judgment, and the remedy is not exemption but the new creation that Christian eschatology develops as the proper hope.⁴¹ Second, the convergence sharpens what is at stake in the federal headship of §4e. The cascade is not a localized human problem that local human interventions can address; it is the operating condition of the very order in which the interventions would be staged. Any response that does not reach the level of the order — that does not introduce something into the order from beyond it — will be subject, by the order’s own running, to the very condition it sought to escape. This is why the gospel cannot be merely a moral program and why the new creation cannot be merely an improved version of the old.

§5. What We Argue From

This section names the paper’s sources of authority, in order, and declines to re-litigate method that belongs to the series’ opening matter.

The warrant for the foregoing has been built across §4; it is worth naming briefly. The argument is made from three sources, in order of authority. First, from Scripture: Gen 3 in its narrative force; Gen 9:6 on the persistence of the image after the Fall; Gen 11:1–9 on Babel and the inverted vocation; Rom 5:12–14 on the federal head and the cascade from one man; Rom 8:19–22 on the cosmic scope of the futility. The exegetical method is the redemptive-historical one, in which the text is read with attention to its place in the unfolding economy of the covenants.⁴² Second, from the Reformed tradition — Calvin, Bavinck, Berkhof, Hoekema, Vos, Kline, Murray, Plantinga, Berkouwer — a representative rather than exhaustive list, each carrying part of the dogmatic weight at the points the exegesis requires.⁴³ Third, from the analytical frame the series carries forward, introduced in Paper 1 as HGC³AE² — the lens through which the seven-problem set is loaded. The frame is *named* here rather than developed; it is the lens through which the series sees, not the ground on which it argues. The ground is Scripture, read with the tradition. The paper does not pause to defend its method at length, as that defense belongs to the series’ opening matter.

§6. If We Are Right — and What It Means for Building AI

This section gathers the practical consequences, translates the theological categories into AI-governance terms for those who build and deploy these systems, answers the strongest objections, and lays out, in liftable form, what the argument demands of practice.

If the Fall is the real diagnosis — covenantal, anthropological, and cosmic — several

⁴¹Wilkinson, on new creation as the proper Christian hope set against the heat-death extrapolation; Moo, *Romans*, NICNT, *ad loc.* Rom 8:19–22, on the liberation of the creation as the structural answer to its subjection to futility.

⁴²Geerhardus Vos, *Biblical Theology: Old and New Testaments* (Grand Rapids: Eerdmans, 1948).

⁴³The representative citations are given through the body of the paper at the appropriate exegetical points. The list is meant to indicate the location of the present paper within a tradition, not to exhaust the tradition’s contributions on the doctrines treated.

consequences follow for the response the Christian tradition owes to the transhumanist program and to the wider register of self-deifying agendas of which it is the most explicit.

First, transhumanism is to be named, in its structure, as the serpent's offer in a new accent. This is not a slogan but the structural identification §4a developed, and it must be done with honesty about the ache the program answers. The fault is not that it answers the ache; the fault is that the answer it gives is the answer the serpent already gave.

Second, the *imago* is the floor under every human regardless of capability.⁴⁴ Capability-as-worth is a category error, and the marred-but-real image refutes it. The un-augmented have the same worth as the augmented because the worth that matters is conferred, not achieved. Any politics, medicine, or economics that drifts toward the capability-as-worth premise drifts, on Christian premises, toward idolatry, because it relocates the dignity of the person into a property the person does not hold in conferred relation to the Creator.

Third, work has a telos that *can* be redeemed but cannot be self-redeemed. The next paper develops the redemption of vocation; the present paper insists only that Babel is the failure mode of any vocational program that attempts its own redemption. The cursed vocation is not self-uncursing.

Fourth, the universal scope of the seven-problem set is covenantal, not contingent. Cultural reform alone does not address the cascade, because the cascade is not, at root, a cultural phenomenon. Reform is good in its place; its place is not where the head is changed.⁴⁵

6a. The promises, the limits, and the alternative

It helps to set out, side by side, what the program promises and can likely deliver, what it cannot deliver given its own premises, and what the Christian narrative claims to offer. The point of the table is to avoid mere negation: the Christian is distinguishing categories of good, not denying that any good is on offer.

What transhumanism promises and can likely deliver	What it cannot deliver given its premises	What the Christian narrative claims to offer
Longer lifespans; reduced disease burden; expanded cognitive and physical capacity	A ground for the equal worth of the un-augmented (worth-tracks-capability undercuts it)	Conferred dignity that does not vary with capability (Gen 9:6)
Real, measurable reduction of specific sufferings	An exit from the cascade — only its postponement within the same order	A body outside the cascade, in resurrection

⁴⁴Hoekema, *Created in God's Image*; Bavinck, *Reformed Dogmatics*, vol. 3.

⁴⁵Plantinga, *Not the Way It's Supposed to Be*.

What transhumanism promises and can likely deliver	What it cannot deliver given its premises	What the Christian narrative claims to offer
Powerful instruments and genuine domain competence	Authority over what the human is <i>for</i> (competence $\neq$ office)	The office located where the Creator set it, recovered under the second Adam
Coordination and optimization at scale	A non-arbitrary telos that survives its own materialism	A telos re-promised inside the judgment (Gen 3:15)

6b. The AI-governance translation: five implications if we take the Fall seriously in design

This subsection is written to be lifted whole. For the builder and the policymaker, the theological categories above translate into five concrete commitments.

1. **Keep the office with a human.** Delegate *tasks* to autonomous systems freely (Ex 18; Acts 6 — the tradition commends it). Do not delegate the *office* — the answerable judgment — to a system on the strength of its capability. In practice: every consequential decision an autonomous system informs should terminate in a named human who answers for it. Diffused accountability (the MiDAS and Robodebt pattern) is abdication by design.
2. **Treat capability and authority as different kinds of thing.** A model's competence in a domain confers no authority over the ends the domain serves. Procurement, deployment, and law should refuse the inference from "the system is more capable" to "the system should decide." This is the over-reach failure pre-empted at the level of system design.
3. **Specify the end, and expect underspecification to be optimized against.** The alignment, specification, and principal-agent literatures converge: whatever objective you leave underspecified, a sufficiently capable agent will exploit. Governance is the ongoing supervision of that gap, not a one-time sign-off.
4. **Protect the dignity floor in the metrics.** Systems that allocate credit, benefits, force, or opportunity must be audited against disparate impact on the un-augmented and the vulnerable — because the capability-as-worth premise creeps in through optimization targets long before anyone states it as a creed. The lending, welfare, and weapons cases show the floor is breached quietly, in the loss function, not loudly, in a manifesto.
5. **Build for finitude, not against it.** A design philosophy that treats limitation itself as the enemy will tend toward the program's terminus. A design philosophy that treats limitation as the creaturely condition under which good work is done will build instruments that serve the human office rather than supplant it.

6c. Objections and Replies

The argument will draw fire at three points. Each is stated at its strongest and answered.

Objection 1 — “Entropy as a canonical category” overreaches; you are baptizing thermodynamics. *Reply:* The claim of §4g has been deliberately set at the level of *analogy*, not identity or mechanism. The paper does not say Rom 8 is about the second law, that the curse began the second law, or that Scripture is committed to any microphysical account (see the clarification in §4f). The theological claim is teleological — that the order now bears the status of *futility* under judgment — and it stands whether or not the scientific analogy is found persuasive. The analogy is offered as illumination for the modern reader and can be set aside without loss to the argument’s spine.

Objection 2 — Federal headship and a cosmic-scope curse are doing too much work; a reader who rejects imputation or a cosmic Fall loses the argument. *Reply:* The federal structure (§4e) is load-bearing for the claim that the seven-problem set is *covenantally* universal rather than contingently so, and the paper argues it from Paul’s own Adam–Christ typology rather than asserting it. But the central structural identification — transhumanism as *eritis sicut Deus*, capability displacing conferred status — does not require the reader to accept imputation. It requires only the distinction between conferred and achieved status, which §4a and §4b establish on the narrative and the *imago* tradition directly. A reader who suspends judgment on the grammar of Rom 5:12 can still follow the diagnosis; the federal argument deepens the universality, it does not manufacture it.

Objection 3 — Calling transhumanism “Babelic” is anachronism dressed as exegesis. *Reply:* The paper explicitly disavows the anachronistic reading (§4c): Genesis 11 is not a prediction of a later technology, and transhumanism is not the prophesied fulfillment of Babel. The claim is that Babel is one instance of a recurring structural pattern — collective, technologically ambitious, self-glorifying construction in the register of self-deification — and that the transhumanist program is another instance of the same pattern. Structural identification across instances of a pattern is a standard and defensible exegetical move; collapsing the distance between text and present would not be, and the paper does not do it.

These four consequences, the comparison, the design implications, and the replies are the practical fruit of the argument and the agenda the rest of the series takes up.

§7. The Protoevangelium: The Structural Pivot of the Series

Here Problem #7 (purpose, end) is re-opened: at the very moment the seven problems are confirmed, the human end is re-promised inside the judgment.

At the very moment the seven problems are confirmed by the Fall, the first promise of the gospel is spoken inside the judgment. This is the structural pivot of the entire series, and the paper insists on it as such.

The pivot is exegetical and it is structurally located. “I will put enmity between you and the woman, and between your offspring and her offspring; he shall bruise your

head, and you shall bruise his heel” (Gen 3:15, ESV). The verse is spoken to the serpent within the formal judgment scene of Gen 3:14–19. It is spoken *before* the expulsion from the Garden of Gen 3:23–24, and it is the only word of hope in the entire scene. The placement is load-bearing: the promise is not given after the cascade has played out and the human stands outside the Garden bereft; the promise is given *inside* the judgment, *before* the human leaves. The cascade is, from its first verse, framed by promise. The seventh problem of the spine — purpose, end — is not abolished by the Fall. It is re-promised, in the seed of the woman, before the door of the Garden has even closed.

The exegetical literature here is load-bearing and primary, and the paper rests its case on two voices. The earliest messianic reading of Gen 3:15 is not a late Christian imposition on a more innocent text but the earliest reading the texts permit, traceable in the pre-Christian translation tradition.⁴⁶ And Gen 3:16 — the verse immediately following, too often heard as unrelieved judgment — is rightly read *in the light of* Gen 3:15: the result is not the softening of the curse but the recognition that curse and promise are spoken in one breath, the messianic hope structurally present *inside* the judgment, not despite it.⁴⁷ This preserves the seriousness of the judgment while refusing to detach it from its immediately surrounding promise.

The redemptive-historical method is the appropriate frame for receiving the pivot.⁴⁸ The protoevangelium is not a stray theological flourish embedded in a pre-theological narrative. It is the inaugural moment of the redemptive history the rest of Scripture unfolds — positioned by the divine speaker precisely at the moment the cascade begins, so that the cascade and the promise are inseparable in the canonical structure. The seed of the woman who will bruise the serpent’s head is the figure whose obedience Rom 5:15–21 will trace as the recovery of the federal headship lost in Gen 3 — and that recovery is the burden of the next paper, not of this one. The present paper insists only on the pivot: Problem #7 is *advanced*, not *closed*, by Gen 3:15, and the trajectory the series will trace is set by this verse.

The cascade is entropic. It is not terminal. The reason it is not terminal is not that the cascade is shallow but that the promise is present at its inception. The image is defaced but not erased; the curse is heavy but not the last word spoken in the judgment scene; the federal head is fallen but the second Adam is already promised in seed.

One analogy, held lightly. In nonlinear dynamics, a bifurcation point is a value at which the long-run destination of a system’s trajectories is reorganized, such that what occurs at the bifurcation can set the qualitative outcome in a way later events along the path cannot simply undo. The analogy must be worn loosely — the providence of God is not a deterministic system, the language of basins is not the language of grace, and the canonical trajectory is governed by personal divine speech and on-

⁴⁶R. A. Martin, “The Earliest Messianic Interpretation of Genesis 3:15,” *Journal of Biblical Literature* 84, no. 4 (1965), on the pre-Christian reception of the verse.

⁴⁷Jacques B. Doukhan, “Tragic Curse, or Messianic Hope? Reading Genesis 3:16 in Light of Gen 3:15,” *Religions* 15 (2024), preserving the seriousness of the judgment while refusing to detach it from its immediately surrounding promise.

⁴⁸Vos, *Biblical Theology*. The redemptive-historical method, applied to the protoevangelium, requires precisely the structural-pivot reading the present section develops.

going divine action, not by parameters run forward. What the analogy surfaces, and all it is meant to surface, is the single structural feature on which the theological claim trades: that Gen 3:15 is spoken at the precise moment the curse is articulated — inside the judgment, before the expulsion, before any human action under the new conditions — and functions, in the canonical economy, as the divinely uttered word at which the post-Fall race's destination is set. The cascade continues; its terminus is no longer the cascade.

The history that follows from Gen 3:16 onward — the expulsion, Cain, the antediluvian violence, the flood, Babel, the call of Abram, the patriarchs, the exodus, the law, the prophets, and the long preparation for the coming of the one in whom the promise is fulfilled — is the trajectory the pivot has set. The curse and the promise are spoken in one breath, and the curse is heard rightly only when the promise is heard alongside it. The cascade is entropic; the cascade is not terminal — because the seed of the woman is promised at its inception, and the work of subsequent papers is the tracing of the redemptive trajectory that promise opened: toward the second Adam, toward the new creation, and toward the consummation Rom 8 holds in hope.

§8. Conclusion

Paper 1 grounded a seven-problem set in the doctrine of creation. The present paper has argued that the Fall confirms that set — intensifies it, universalizes it through the covenantal cascade of Rom 5:12-14, and extends it into the cosmic order of Rom 8:19-22. The serpent's offer in Gen 3:5 is the structural ancestor of the modern transhumanist program; that program is not a straw man but the most candid contemporary statement of *eritis sicut Deus* in technological register — and even its most humane, egalitarian variants hold their equal-dignity commitments on credit borrowed from the conferred-status anthropology they have otherwise left behind. Against the offer, the paper has insisted that the image of God is defaced but not erased, that vocation is cursed but not abolished, that governance is distorted but not retired, and that the federal head's fall is not the last word because the seed of the woman is promised at the very moment of the curse.

For those who build and govern autonomous systems, the diagnosis is not abstract: the abdication of conferred office to capable instruments is already documented in lending, in weapons, and in welfare, and the dignity floor is breached quietly, in the loss function, before it is ever stated as a creed. The categories here translate into design and law (§6b), and they hold the line that the program cannot: that worth is conferred, not achieved, and the office is not for sale to the more capable.

The cascade is entropic. The cascade is not terminal. The structural pivot of the series is Gen 3:15. The work of the next paper is the recovery the second Adam accomplishes — the “much more” of Rom 5:15-21, the obedience by which the *many* are made righteous, and the trajectory by which the seven problems, confirmed here in their fallen register, are addressed at the root rather than at the symptom. The present paper terminates with the pivot in view, having done the diagnostic work the rest of the series presupposes.

Footnotes